Western Governor's University

D205 Data Acquisition

Performance Assessment

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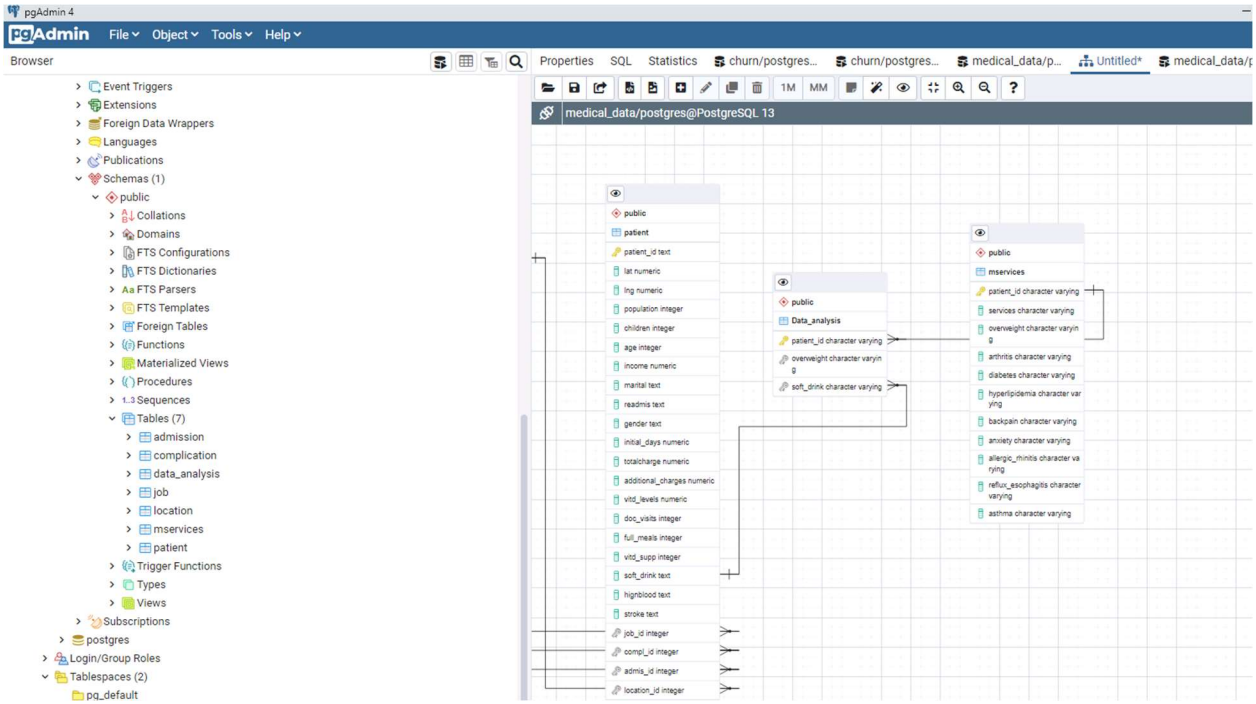
8/1/2024

A) How many overweight patients consume drinks relative to the total number of overweight patients?

1) The overall number of overweight patients who consume soft drinks must be compared to the number of total overweight patients. An intermediate table called 'mservices' will be used to hold all data from the mservices.csv file to allow for easy transfer of columns. Now, the table 'patient' and new table 'mservices' must be directly compared by creating a new table called data_analysis and importing the columns patient_id, overweight and soft_drink. Then, the total counts will be calculated from the data_analysis table by using SQL's COUNT function for both the overweight and soft_drink column, using where conditions. This will flow into a final table to view the results.

2) The patient ID field will be used to inner join the patient table and the mservices.csv file.
Then, the 'overweight' text column from patient will be imported, as well as the column soft_drink from the mservices table. This will be used to create a new table data_analysis consisting of patient_id as varchar, and overweight/soft drink consumption columns as varchars. The COUNT function using where conditions will be inserted into an answers table for viewing the total amount of overweight patients, and the number of overweight patients who consume soft drink.

B) ERD:



Patient_id has M:N relationships with both the columns overweight and soft_drink, as patient IDs can be linked to multiple medical conditions or survey responses, and each condition/response can be linked to multiple patient IDs.

1) The newly created data_analysis table will be displaying data imported from the patient table and mservices table, using the column patient_id as the primary key. The patient_id and 'overweight' columns will be imported into data_analysis from 'patient' table. Mservices soft_drink will be referenced for the soft_drink column. Potential issues with joining these tables would be a lack of complete data resulting in NULL values in either the overweight or soft_drink columns being imported, as this information may not be available for each and every patient. This could skew results if null values are present.

2)

```
CREATE TABLE mservices(
        Patient_id VARCHAR,
        Services VARCHAR,
        Overweight VARCHAR,
        Arthritis VARCHAR,
```

```
Diabetes VARCHAR,
Hyperlipidemia VARCHAR,
BackPain VARCHAR,
Anxiety VARCHAR,
Allergic_rhinitis VARCHAR,
Reflux_esophagitis VARCHAR,
Asthma VARCHAR,
PRIMARY KEY(patient_id)
```

3)
```
COPY public.mservices (patient_id, services, overweight, arthritis, diabetes, hyperlipidemia,
backpain, anxiety, allergic_rhinitis, reflux_esophagitis, asthma)
FROM 'C:\LabFiles\Medical\Mservices.CSV'
DELIMITER ','
CSV HEADER ;
```

C)

```
CREATE TABLE data_analysis
AS (select patient_id, soft_drink
FROM patient)
```

```
CREATE TABLE calc1
 AS
SELECT data_analysis.patient_id, m.overweight, data_analysis.soft_drink
FROM mservices AS m
INNER JOIN data_analysis ON data_analysis.patient_id = m.patient_id ;
```

```
CREATE TABLE final_table
(total_over INT,
Overandsoftdrink INT) ;
```

```
INSERT INTO final_table (total_over)
SELECT COUNT(*)
FROM calc1
WHERE overweight ='Yes' ;
```

```
INSERT INTO final_table(overandsoftdrink)
SELECT COUNT(*)
FROM calc1
WHERE overweight ='Yes' AND soft_drink = 'Yes'
```

1) The exported data file is in the LabsOnDemand 'download' folder labeled as BrandonVelezD206PA.csv.

D) The information in the patient table and mservives.csv file would reasonably be refreshed on a quarter annual basis. This will inform decision makers of relevant timely data, while excluding irrelevant intermediate fluctuations.

1) Quarter annually is the basis chosen to refresh this data, as the long-term shifts are relevant to business decisions, while tracking short term fluctuation is not worth the extra resources. This can track the correlation over time in response to any changes, such as a new soft drink release, and to track general trends over long periods of time. Refreshing monthly or weekly is not believed to be an efficient use of resources, as soft drink consumers are not expected to shift between consumption and not consuming often. It is also not likely that patients will often fluctuate back and forth between obesity and being below the overweight standard within three-month periods.

E) Panopto Link:

https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=fddd6eb0-d767-4d6c-a16d-b1c4016bd91f

F) **Sources**

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